



Atmospheric and Delivered Image Quality in OpSim

Bo Xin, George Angeli, and Željko Ivezić

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1. Introduction

This technical note describes the algorithms used in OpSim to calculate atmospheric and delivered image quality. As opposed to atmospheric seeing, atmospheric image quality ($FWHM_{atm}$) is the limit achievable with a perfect telescope-instrument combination through the given atmosphere. It is obtained by scaling the zenith pointing atmospheric image quality at 500nm ($FWHM_{500}$) with appropriate wavelength and airmass. The delivered image quality, characterized by the physical $FWHM$ ($FWHM_{tot}$) and the effective $FWHM$ ($FWHM_{eff}$) of the delivered PSF , is obtained by combining the atmosphere and the system contributions.

The effective $FWHM$ is directly related to the 5σ detection limit, m_5 . Its definition can be found in LSE-40. Background information and more detailed reasoning to support the particular atmospheric model used can be found in Document-18208. The appendix of Document-18208 also outlines how to compute $FWHM_{atm}$ starting from $r_0(500nm,zenith)$ by including the outer scale correction factor, and how to obtain $FWHM_{tot}$ and $FWHM_{eff}$ properly by convolving the von Karman atmosphere with the system contribution. This technical note presents simplified formulae for those computations, which are more suited for use by OpSim.

2. Inputs and Outputs

2.1. Inputs

Reference atmospheric image quality ($FWHM_{500}$) is the $FWHM$ of a point source (PSF) possibly detected by a perfect instrument at 500 nm wavelength and zenith pointing, through the



atmosphere on Cerro Pachon. The atmospheric refraction index fluctuations are assumed to follow the von Kármán model. A 10-year time history of atmospheric image quality is currently available in OpSim. In this document, *FWHM is measured in arcsec*, unless otherwise noted.

System image quality ($FWHM_{sys}$) is the specified i.e. required image quality the LSST system is expected to deliver at 500 nm wavelength and zenith pointing. By the Observatory System Specification (LSE-30), under these conditions the delivered image quality is 0.4 arcsec ($FWHM_{sys0}$).

Outer scale (L_0) of the von Kármán atmospheric model characterizes the correlation length in the atmosphere. The expected outer scale at the LSST site is 30 meter. In this document, we consider L_0 not a variable by always using $L_0=30m$.

Wavelength (λ) and zenith angle (z) dependence of delivered image quality is included in the model described here. In this document, *wavelength is measured in nm*, unless otherwise noted. For all analysis described in this note, we use the effective wavelength in each of the 6 optical bands. The values are $\lambda_u=365.0$, $\lambda_g=480.0$, $\lambda_r=622.0$, $\lambda_i=754.0$, $\lambda_z=868.0$, and $\lambda_y=973.0$.

2.2. Outputs

Effective size of delivered image ($FWHM_{eff}$) represents the optimal sky area (n_{eff} in pixel count) around an unresolved object to be included in photometry (LSE-40). $FWHM_{eff}$ is directly related to the limiting depth m_5 , and defined as,

$$FWHM_{eff} = 0.663 \text{ pixelScale} \sqrt{n_{eff}} \quad (2.1)$$

Physical size of delivered image ($FWHM_{tot}$) represents the combined effect (convolution) of the atmospheric and system $PSFs$. There are applications, such as star/galaxy performance metric and trailing losses for moving objects that require the knowledge of the geometric $FWHM$ of the delivered image.

3. Algorithms

3.1. Calculating atmospheric image quality

By definition, atmospheric seeing is the function of r_0 and the wavelength λ . Following the usual convention, r_0 is measured in cm, unless otherwise noted.

$$FWHM_0(\lambda, z) = 0.976 \frac{\lambda}{r_0(\lambda, z)} (2.063 \times 10^{-2}) \quad (3.1)$$

As stated in Sec. 2.1, $FWHM_0$ is in unit of arcsec, and λ in unit of nanometer.

The dependence of r_0 on wavelength and zenith angle is well established¹. (Here $X = \sec(z)$ is the air mass.) Consequently, atmospheric seeing scales with wavelength and air mass in a straightforward way.

$$\begin{aligned} r_0(\lambda, z) &= r_0(500, 0) \left(\frac{\lambda}{500} \right)^{\frac{6}{5}} [\cos(z)]^{\frac{3}{5}} = r_0(500, 0) \left(\frac{\lambda}{500} \right)^{\frac{6}{5}} X^{\frac{3}{5}} \\ FWHM_0 &= FWHM_0(500, 0) \left(\frac{\lambda}{500} \right)^{-\frac{1}{5}} [\cos(z)]^{\frac{3}{5}} \end{aligned} \quad (3.2)$$

Atmospheric image quality² also takes into account the outer scale, besides r_0 .

$$FWHM_{atm}(\lambda, z) = 0.976 (2.063 \times 10^{-2}) \frac{\lambda}{r_0(\lambda, z)} \sqrt{1 - 2.183 \left(\frac{r_0(\lambda, z)}{L_0} \right)^{0.356}} \quad (3.3)$$

There have been two proposals on how we calculate $FWHM_{atm}$ in OpSim, as outlined in the following two subsections. OpSim has adopted the second option.

3.1.1. Using r_0 as the key in computing $FWHM_{atm}$

The first way to compute $FWHM_{atm}$ is very straightforward and fast. It involves the following steps.

1. Reconstruct $r_0(500, X=1)$ time history from $FWHM500$, using a LUT generated from the special case of Equation (3.3) below. Please note, this operation needs to be done only once, outside of the OpSim kernel.

$$FWHM500 = 0.976 (2.063 \times 10^{-2}) \frac{500}{r_0(500, z)} \sqrt{1 - 2.183 \left(\frac{r_0(500, z)}{30 \times 10^2} \right)^{0.356}} \quad (3.4)$$

Figure 1 shows $FWHM500$ as a function of $r_0(500, X=1)$, in the range [10, 20]cm.

2. Using Equation (3.2), scale r_0 with wavelength and zenith angle.
3. Using the scaled r_0 and Equation (3.3), calculate atmospheric image quality $FWHM_{atm}$.

¹ F. Roddier, *Adaptive Optics in Astronomy*, Cambridge University Press, **1999**

² See Equation (19) of Tokovinin, A. *From Differential Image Motion to Seeing*, PASP, 114, 1156, **2002**.

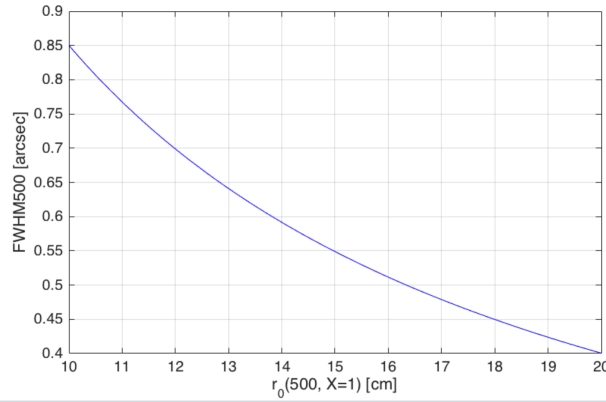


Figure 1 The relationship between $FWHM_{500}$ and $r_0(500, X=1)$ as defined by Equation Error!
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3.1.2. Relating $FWHM_{atm}$ to $FWHM_{500}$ with r_0 behind-the-scene

The second way to go from $FWHM_{500}$ to $FWHM_{atm}$ is to create a grid of data in the 3D space $(r_0(500, X=1), \lambda, X)$, where $X=1/\cos(z)$ is the airmass. Within this 3D space, we can calculate $FWHM_{500}$ using Equation (3.4), and $FWHM_{atm}$ using Equation (3.3). Because $FWHM_{500}$ is only a function of $r_0(500, X=1)$, the independent variables can as well be $(FWHM_{500}, \lambda, X)$. We fit the $FWHM_{atm}$ “data” calculated from Equation (3.3) to the following functional form,

$$FWHM_{atm} = FWHM_{500} \left(\frac{\lambda}{500} \right)^\alpha X^\beta \quad (3.5)$$

With $FWHM_{500}$ in (0.3, 1.5) arcsec, wavelength being the effective wavelength in the 6 optical bands, and zenith angle between 0 and 60 deg., a nonlinear least square fit yields $\alpha = -0.2880$ and $\beta = 0.6372$, with residual RMS of 7 mas and PV of 62 mas. Compared to the wavelength and airmass dependence of the atmospheric seeing (Equation (3.2)), where $\alpha = -0.20$ and $\beta = 0.60$, $FWHM_{atm}$ gets better faster when wavelength increases, and deteriorates faster with larger airmass.

To visualize the difference in these power parameters, we project the data onto wavelength and airmass separately. Figure 2 shows the log-log plots of $FWHM_{atm}/FWHM_{atm}(500, X)$ vs. wavelength and $FWHM_{atm}/FWHM_{atm}(\lambda, X=1)$ vs. airmass. The blue circles are “data” as described above. The blue lines are results of the fit. For comparison, the red lines show standard values for the atmospheric seeing: $\alpha = -0.20$ and $\beta = 0.60$. For the upper plot, the spread of the data in vertical direction is due to the spread in $FWHM_{500}$ and airmass. For the lower plot, the spread in the data in the vertical direction is due to the spread in $FWHM_{500}$ and wavelength.

From Figure 2, we see that blue circles, therefore also the new α value, deviate from -0.2 quite significantly. The new β , however, is not very different from 0.6. In the OpSim algorithm, we use Equation (3.5) with rounded values for α and β : $\alpha = -0.3$ and $\beta = 0.6$. These values lead to a residual RMS of 16 mas and PV of 104 mas, which are still acceptable.

Both ways of calculating $FWHM_{atm}$ are computationally inexpensive. The first way, which involves r_0 , is more accurate and more intuitive for the community familiar with imaging through atmosphere. However, considering that OpSim already has a ten-year history of $FWHM_{500}$, and

Equation (3.5) provides a single, simple form that is familiar to astronomers, the second way of getting $FWHM_{atm}$, has been adopted for OpSim use.

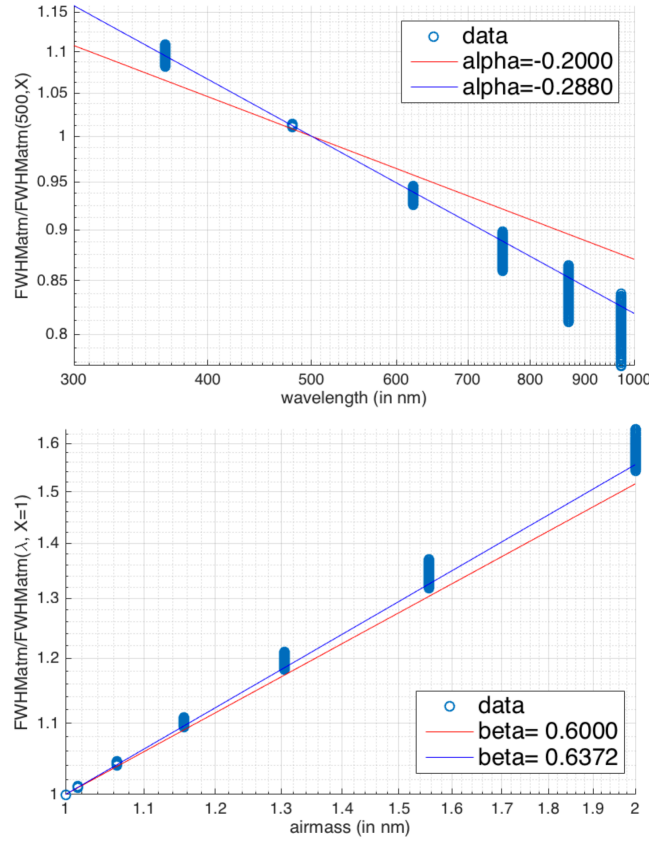


Figure 2 Log-log plots of $FWHM_{atm} / FWHM_{atm}(500, X)$ vs. wavelength (top) and $FWHM_{atm} / FWHM_{atm}(\lambda, X=1)$ vs. airmass (bottom).

3.2. Calculating system image quality

Per the Observatory System Specification (LSE-30), the system image quality is allowed to degrade with increasing zenith angle.

$$FWHM_{sys} = FWHM_{sys0} [\sec(z)]^{3/5} = FWHM_{sys0} \frac{1}{[\cos(z)]^{3/5}} \quad (3.6)$$

It is assumed to have a Gaussian PSF , as it is (i) the sum of large number of statistically independent effects, and (ii) the major contributor (charge diffusion in the CCD) is truly Gaussian.

The current system image quality requirement in the OSS is:

- Camera 0.3", Telescope 0.25", system design 0.08"; all added together in quadrature equals to 0.4".

- It is allowed to degrade with zenith angle as $\sec(z)^{0.6}$. It is following the image degradation due to airmass, but specified for the observatory optical system, separately from the air contribution.

The system $FWHM$ is therefore calculated as

$$FWHM_{sys}(X) = X^{0.6} \sqrt{0.25^2 + 0.30^2 + 0.08^2} \quad (3.7)$$

3.3. Combining atmospheric and system image quality

$FWHM_{tot}$ is the characteristics of the convolution of atmospheric and system $PSFs$. Furthermore, calculating $FWHM_{eff}$ involves the square integral of the combined PSF . Both of these calculations are more computationally intensive than it is appropriate for the OpSim kernel. Consequently, using simplified fits in OpSim is desirable.

We then need to come up with an empirical function for relating $FWHM_{eff}$ and $FWHM_{tot}$ to $FWHM_{atm}$ and $FWHM_{sys}$. Considering that in the case when both the atmospheric and system $PSFs$ are Gaussian, the $FWHM$ of the convolved profile would simply be the RSS of $FWHM_{atm}$ and $FWHM_{sys}$, we use the following function,

$$FWHM_{eff} = c \sqrt{FWHM_{sys}^2 + d \times FWHM_{atm}^2} \quad (3.8)^3$$

$$FWHM_{tot} = e \times FWHM_{eff} + f \quad (3.9)^4$$

To generate the “data” that we can use to fit to the above parameters (c , d , e , and f), we first pick a set of $r_0(500, X=1)$ values that would give $FWHM_{atm}$ in the range of (0.3, 1.5) arcsec. After scaling $r_0(500, X=1)$ with appropriate effective wavelength in each band, von Karman structure functions are generated⁵, which are then used to calculate the OTF. Airmass variation is ignored here because its effect on the shape of the atmospheric PSF is equivalent to a change in $r_0(500, X=1)$.

The system contribution is represented by a Gaussian profile with $FWHM$ between 0.35 and 0.55 arcsec. The OTF of the system PSF and the OTF of the atmosphere are then multiplied together to get the combined OTF. The convolved PSF profile is obtained as the inverse FFT of the resulting OTF. $FWHM_{eff}$ and $FWHM_{tot}$ are then numerically calculated.

³ One subtlety here is that strictly speaking $FWHM_{eff}$ is also a function of wavelength. As can be seen from Equation (3.3) **Error! Reference source not found.**, $FWHM_{atm}$ is a function of $r_0(\lambda, z)$ and λ , or, to make the variables independent, a function of $r_0' = r_0(500, X=1) * X^{0.6}$ and λ . Providing a value for $FWHM_{atm}$ doesn't uniquely determine r_0' , therefore is not enough for generating the shape of the atmospheric PSF . The wavelength also needs to be provided. In the initial modeling, we also included a factor $(\lambda/500)^a$ after the coefficient d in Equation (3.8). That turned out to have minimal impact on the residual. So we decided to leave it out.

⁴ Equation (3.9) **Error! Reference source not found.** assumes that $FWHM_{tot}$ and $FWHM_{eff}$ are sufficiently correlated. Initially we used the same functional form for $FWHM_{tot}$ as in Equation (3.8). That didn't help much with the fit residual. We therefore use Equation (3.9) because it is simpler.

⁵ See Equation (18) of Tokovinin, A. *From Differential Image Motion to Seeing*, PASP, 114, 1156, 2002.

The fit of the data to Equation (3.8) yields $c=1.16$ and $d=1.04$, with residual RMS of 12 mas, with PV of 55 mas. Figure 3 shows the residual of the fit. The fit to Equation (3.9) yields $e=0.822$ and $f=0.052$, with residual RMS of 15 mas, with PV of 67 mas. Figure 4 shows the result of the fit. It is seen that both functions describe the data very well.

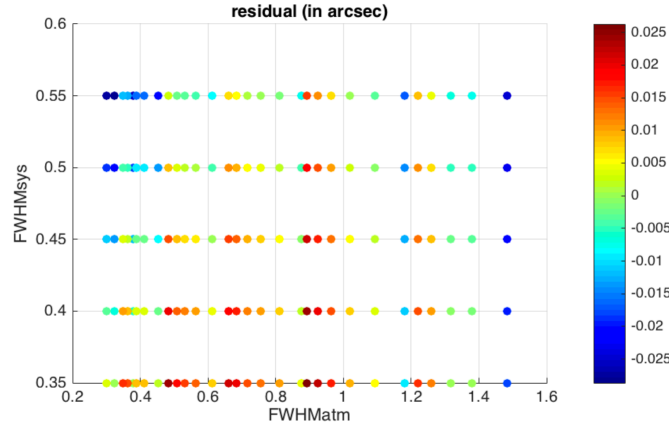


Figure 3 Residual of the fit to Equation (3.8) The x- and y-axes are the two variables: $FWHM_{atm}$ and $FWHM_{sys}$. The color of the points indicates the value of the residual.

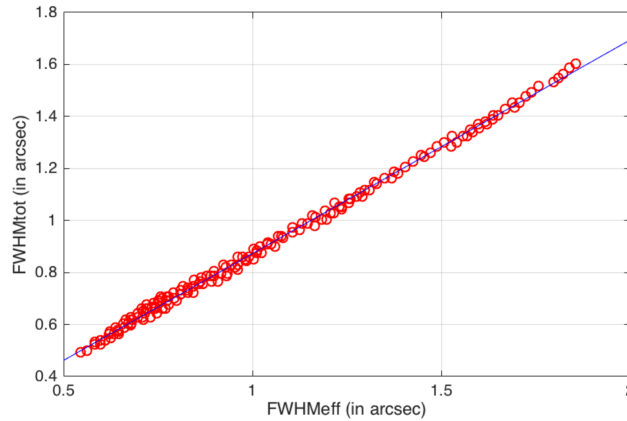


Figure 4 $FWHM_{tot}$ as a linear function of $FWHM_{eff}$. Red circles show data points. The blue line shows the result of the fit to Equation (3.9).

4. Summary

The algorithm in OpSim for calculating the atmospheric and delivered image quality is summarized as follows.

$FWHM_{atm}$ is a function of $FWHM500$, wavelength λ , and airmass X ,

$$FWHM_{atm} = FWHM500 \left(\frac{\lambda}{500} \right)^{-0.3} X^{0.6} \quad (4.1)$$

$FWHM_{sys}$ is a function of airmass X only,



$$FWHM_{\text{sys}} = X^{0.6} \sqrt{0.25^2 + 0.30^2 + 0.08^2} \quad (4.2)$$

Once we have $FWHM_{\text{atm}}$ and $FWHM_{\text{sys}}$, the delivered $FWHM_{\text{eff}}$ and $FWHM_{\text{tot}}$ can be obtained via

$$FWHM_{\text{eff}} = 1.16 \sqrt{FWHM_{\text{sys}}^2 + 1.04 FWHM_{\text{atm}}^2} \quad (4.3)$$

and

$$FWHM_{\text{tot}} = 0.822 FWHM_{\text{eff}} + 0.052 \quad (4.4)$$